

A-level

ENVIRONMENTAL SCIENCE

Paper 2 - Generated practice paper Set 30

Mark scheme

120 marks: Questions 1-10 (95 marks) plus one essay from Question 11 (25 marks).

Marking guidance

- Marking points are separated by bullets; each bullet is worth 1 mark unless stated.
- Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
- Alternative correct wording and other relevant, accurate points are credited.
- For levels-of-response questions, first determine the level using the descriptors, then position the mark within the level using the indicative content.
- Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	• Size: shows whether the population is viable / how urgent action is; sets a baseline to judge success; needed to estimate carrying capacity and sustainable off-take • Distribution: shows which habitats / areas are used and whether the population is fragmented into non-viable isolated groups needing corridors; identifies key areas to protect • Survival rate (mortality): identifies the life stage and causes of death (predation, disease, poaching) so management targets the limiting factor • Age structure: many young indicates a growing population; few young / ageing population indicates breeding failure or high juvenile mortality even if adults are numerous; k-selected species need breeding adults protected	8
01.2	• Direct counts from vehicles / aircraft / drones along transects, or camera traps with individual recognition • Sub-sample counts in known areas scaled up to the whole reserve; mark-release-recapture (Lincoln index) for smaller species	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 393-394, 400-402; AQA spec 3.1.2.3.5.	
02.1	• Warming melts ice shelves that hold back glaciers and lubricates ice flow so land ice reaches the sea faster (Antarctic Peninsula most affected); sea ice loss removes krill habitat and lowers albedo • Raised UV harms planktonic algae, krill and fish / sea urchin larvae in surface waters (land animals protected by fur / feathers) • Disturbance of wildlife, introduction of non-indigenous species and pathogens, fuel and sewage pollution; visitor numbers rising rapidly at a few landing sites • Reserves of coal, oil, gold, uranium, platinum etc; pressure to exploit will rise as reserves elsewhere deplete though currently banned and operations extremely difficult	4

Question	Marking guidance	Marks
02.2	• Treaty signed by 52 countries including all active there; no military activity (except supporting research), no nuclear explosions or radioactive waste dumping • No commercial mineral exploitation • Fisheries regulated by CCAMLR: krill and some fish exploited, fisheries closed if over-exploited; no commercial exploitation of mammals • Tourism control: no accommodation on land (wastes stay on ships), one ship per site with maximum 100 ashore, trained guides, keep 5 m from wildlife, no litter, cleaned protective clothing to prevent introductions • Waste management: all wastes removed (fuel containers, batteries, plastics, oil); sewage only discharged where sea currents disperse it; old waste removed unless removal causes more damage • Only long-term residents are research scientists and support staff; strict regulation of research impacts	6
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 69-71; AQA spec 3.1.2.3.4.	
03.1	• Carnivores fed pellets of sandeels, herring, anchovy; overfishing sandeels reduces puffins and terns; reduced by more plant protein / vegetable oil in pellets or farming herbivores (tilapia) • High density spreads lice (which spread to wild fish) and infections; pesticides (organophosphates, pyrethroids, hydrogen peroxide) kill wildlife; antibiotics breed resistant bacteria; reduced by lower densities, single-age tanks with flow from young to old, wrasse as biological control, warm-water washes, circulating flow • Escaped farmed fish breed with wild fish introducing disadvantageous genes (bright colour), or non-indigenous escapees colonise as predators / competitors; reduced by better cage design, sterile triploid fish (pressure-treated eggs), species that cannot breed locally (Japanese oysters in UK) • Faeces and uneaten food decompose causing deoxygenation under cages; reduced by monitored feeding, siting cages in strong currents, effluent treatment / reed beds for freshwater farms, IMTA with seaweed and shellfish absorbing wastes	8
03.2	• Predators are killed under licence to protect stock, reducing wild populations of protected / valued species • Better cage designs, netting, fencing, bird scarers exclude predators without killing	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 356-359; AQA spec 3.5.2.3.	
04.1	• South America: high loss ($5.1-5.5 \text{ Mha yr}^{-1}$) falling by about half by 2020 • Africa: loss steady at about 3.4 then rising to 3.9 by 2020 - now the highest • Europe: negative values - forest area increasing (afforestation exceeds loss), though the gain is shrinking	3
04.2	• South America: clearance for commercial cash crops (sugar cane, soya) and cattle ranching for export; HEP reservoirs; roads improving access; mining • Africa: growing rural populations clearing land for subsistence farming; fuelwood and charcoal collection above regrowth (charcoal wastes much of the wood's energy); fodder cutting in dry areas • Both: hardwood timber (teak, mahogany) from clear-felled mixed forest often not replanted; urban expansion	4

Question	Marking guidance	Marks
04.3	• They create barriers fragmenting wildlife populations • Improved access makes it easier to remove timber and farm produce, so clearance spreads along them	2
04.4	• Conifers grow quickly so plantations are replanted after harvest and yield a saleable crop within decades; slow-growing hardwoods (teak, mahogany) take too long for commercial plantations so they are taken from natural forest that is clear-felled and not replanted	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 367-368; AQA spec 3.5.3.3.	
05.1	• Cross-bred offspring are healthier / more productive because the problems caused by homozygous recessive genes in inbred varieties are reduced (heterosis)	1
05.2	• Zebu tolerate heat but have a low milk yield; Ayrshire have a high milk yield but are not heat tolerant • Cross-breeding combined the desirable characteristics: heat tolerance with a high milk yield	2
05.3	<i>Any 2 from:</i> • Replacement of valuable animals that die with genetically identical individuals • Replacement of herds culled during a disease outbreak • Production of large numbers of individuals with desirable characteristics	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 321-322; AQA spec 3.5.1.2.	
06.1	• Photosynthesis and aerobic respiration balance O₂ and CO₂ in a dynamic equilibrium; evapotranspiration adds water vapour forming clouds and rain; microbes drive the nitrogen and other biogeochemical cycles so wastes do not accumulate and nutrients are recycled • Detritivores, fungi and bacteria decompose dead organic matter releasing nutrients and forming humus that binds soil; organic acids weather rock forming new soil; roots bind soil, reducing erosion and regulating water flow / flooding • Species provide food, pollination (bees, moths, birds, bats - more reliable than wind, especially in forests), seed dispersal (animals carry larger seeds to suitable sites, faeces as fertiliser, gut passage stimulates germination) and habitat provision (trees as nest sites and canopy microclimate; mollusc shells for hermit crabs)	6
06.2	• Its nectar is at the end of a 30 cm tube reached only by the long tongue of the sphinx moth, which carries pollen only between orchids of the same species • Advantage: no pollen wasted on other species and the moth has an exclusive food source; risk: if either species declines the other cannot survive	2
06.3	• They are not ruminants so seeds are not destroyed by repeated chewing / pass through intact; they eat the fruit of most tree species • Not territorial so they carry seeds over large distances / areas	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 18-19; AQA spec 3.1.2.1.3.	

Question	Marking guidance	Marks
07.1	• Ground surface: pitfall trap (container sunk level with the surface, cover, checked after 24 h) - only mobile surface animals; trapped predators eat others; preservative attracts / repels; escapes • Leaf litter: Tullgren funnel (sample on mesh under a light; heat / light drive invertebrates down into a container for a standard time) - only species repelled by heat / light; some die or stop moving; large ones cannot pass the mesh • Branches: beating tray (branch beaten a set number of times over a sheet) - hard to standardise vigour; some species not dislodged / fly away; high branches unreachable	6
07.2	• Moths only fly as adults for part of the year and activity depends on weather / season, so a species may be present but not caught • Brightness / wavelength of the lamp affects attractiveness so catches vary between traps	2
07.3	• The area of ground covered by the suction pipe is known so invertebrates per m² can be calculated • A pooter is only a handling / collection device for individuals already found, not a sampling method	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 397-399; AQA spec 3.7.2.1.	
08.1	• The new opportunity is overwhelmingly attractive (economic benefit) • Lack of knowledge / understanding of the impacts • The impacts occur elsewhere or much later, so are not evident to those who benefit	3
08.2	• Cultural belief that resources should be exploited to satisfy needs only • Problems of over-exploitation become obvious quickly as local supplies dwindle, so behaviour changes	2
08.3	• Resources are brought in from other areas • If one area is depleted, exploitation moves to another area (eg timber, fish, metals, energy) without the consumers noticing	2
08.4	• Easier access to energy has made many new activities possible • The scale of activities has greatly increased • Toxic wastes are produced that cannot be decontaminated by natural processes	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 370-371; AQA spec 3.6.1.	
09.1	• Direct measurements of past conditions cannot be taken, so indirect evidence that is related to the condition must be used	1
09.2	• Age / date of the sample (radioisotope dating) • Past temperatures • Composition of the past atmosphere (eg CO₂ / methane concentrations) • Temperature: more magnesium is incorporated into calcite at higher temperatures	4
09.3	• Movement of continents changed the position of land masses / latitude and therefore the climate and habitats present • Ocean currents redistribute heat and nutrients; changes in currents (eg as continents moved) altered climates, so past conditions cannot be explained without knowing how the systems worked	2

Question	Marking guidance	Marks
09.4	<i>Any 3 from:</i> • There was no oxygen / ozone layer so high levels of UV radiation reached the land surface; water absorbed the UV and protected organisms in the oceans • Warm temperatures around the vents and a rich mix of chemicals / increasingly complex organic molecules in the sea made biological reactions more likely • The atmosphere contained toxic gases such as ammonia; the early Archaea were anaerobic so did not need oxygen • Water acted as the solvent for reactions / transport	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 6-9; AQA spec 3.1.1.1-3.1.1.3.	
10.1	Levels of response Level 3 (7-9 marks): Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. Level 2 (4-6 marks): Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. Level 1 (1-3 marks): Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. 0 marks: No relevant content. Indicative content (credit other relevant, accurate points) • In-situ: species remain in natural conditions with the whole community and inter-species relationships; protected areas (SSSI, NNR, SAC, SPA, Ramsar, MCZ), land ownership, agri-environment schemes, habitat creation and design (area, shape, corridors, diversity), management of abiotic and biotic factors, plagioclimax maintenance, population control, reintroductions and rewilding (Yellowstone wolves, Scottish beavers) • Limits of in-situ: external threats (pollution, climate change), enforcement, cost, conflicts with land use, small fragmented populations • Ex-situ: captive breeding when in-situ is impossible or insufficient; selection criteria; difficulties keeping and breeding (habitat size, food, relationships, breeding triggers, group dynamics, gene pool / inbreeding, hybridisation); techniques (stud books, cryopreservation, AI, embryo transfer, micropropagation, cloning); seed banks; successes (Hawaiian goose, Iberian lynx, red kite, large blue) • Limits of ex-situ: expensive, few species, loss of wild behaviour, small gene pools, release requires suitable habitat and post-release support (hard vs soft release, monitoring) • Judgement: ex-situ is a temporary safeguard / support that only succeeds if in-situ habitat exists to return to; both needed, in-situ fundamental	9
10.2	• Restoring natural processes / communities to a habitat changed by humans, often by reintroducing missing (keystone) species and removing human management such as grazing	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 32-49; AQA spec 3.1.2.3.3-3.1.2.3.4.	

Question	Marking guidance	Marks
11.1	Natural systems have remained sustainable over very long periods. Discuss how the features of natural systems could be copied to make human systems more sustainable. Levels of response Level 5 (21-25 marks): Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. Level 4 (16-20 marks): Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. Level 3 (11-15 marks): Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. Level 2 (6-10 marks): Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. Level 1 (1-5 marks): Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. 0 marks: No relevant content. Indicative content (students are not expected to cover all of these; credit other relevant material) • Dynamic equilibria and negative feedback in climate, hydrological cycle and populations; positive feedback and tipping points caused by humans • Energy: solar-driven, low energy density, low temperature (photosynthesis, enzymes, nitrogen fixation vs Haber process); renewable energy, heat pumps, bioleaching, fuel cells • Materials: cycles, few abundant elements, biodegradable non-toxic wastes vs fossil fuels and dispersed minerals; recycling, Cradle to Cradle, industrial symbiosis • Diversity and stability: rainforests / reefs vs agroecosystems; diverse energy mixes • Optimum vs maximum production; supporting ecosystems in agriculture, forestry, fisheries, urban design • Case studies and metrics; limits and judgement Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 370-379; AQA spec 3.6.1-3.6.4.	25

Question	Marking guidance	Marks
11.2	Discuss the methods that can be used to conserve biodiversity and evaluate their effectiveness. Levels of response Level 5 (21-25 marks): Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. Level 4 (16-20 marks): Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. Level 3 (11-15 marks): Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. Level 2 (6-10 marks): Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. Level 1 (1-5 marks): Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. 0 marks: No relevant content. Indicative content (students are not expected to cover all of these; credit other relevant material) • Setting priorities: IUCN Red List criteria, EDGE, endemic, keystone, flagship species, population dispersal • Legislation and protocols: Wildlife and Countryside Act, designated areas (SSSI to MCZ), CITES Appendices, IWC, EU CFP, ITTO - successes and limits of enforcement • Captive breeding and release: selection, difficulties, techniques, release site criteria, hard / soft release, monitoring; seed banks • Habitat conservation: creation, design (area, shape, corridors, age structure, colonisation), abiotic and biotic management, plagioclimax maintenance, population control, reintroductions • Habitat case studies (woodland, rainforest, coral reefs, islands, mangroves, Antarctica) illustrating threats and conservation efforts • Monitoring and new technologies informing decisions • Evaluation: examples of success (Iberian lynx, Hawaiian goose, Scilly rats, Yellowstone) and failure (ITTO, tigers); cost, politics, climate change; judgement on the most effective combination Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 26-71; AQA spec 3.1.2.3.	25